Quantum Effects in the Phase-State Parameter Cosmological Model (PSP)

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Abstract

This work presents a quantum generalization of the Phase-State Parameter (PSP) cosmological model. We introduce quantization of the phase space, derive the Schrödinger equation for the phase wave function, analyze quantum barriers at the assembly and inversion points of the cycle, investigate tunneling between cycles, and formulate the uncertainty relation for the phase and phase flow. It is shown that quantum effects naturally complement the classical PSP model, providing its fundamentality and connection to observed quantum fluctuations of the cosmic microwave background. The model predicts a discrete phase spectrum, a fundamental limit on calibration accuracy, and the possibility of tunneling between adjacent cycles.

Keywords

quantum cosmology, cycle phase, quantization, Schrödinger equation, tunneling, uncertainty relation, cosmic microwave background

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98.80.-k, 98.80.Es, 95.36.+x, 95.35.+d, 04.60.-m

1 Introduction

The classical PSP model represents a complete, self-consistent, and empirically grounded alternative to ΛCDM. However, any fundamental theory must have a quantum limit. This work proposes a quantum generalization of the PSP model.

Figure 1 presents the potential $V(M)$ of the PSP model, demonstrating quantum barriers at $M = 0$, $M = 0.5$, and $M = 1$.

2 Quantization of Phase Space

In the classical PSP model, the phase $M \in [0, 1]$ is continuous. In the quantum version, we introduce the Bohr-Sommerfeld quantization condition:

$$\oint p_M \, dM = nh, \quad n \in \mathbb{N}, \tag{1}$$

which yields a discrete spectrum:

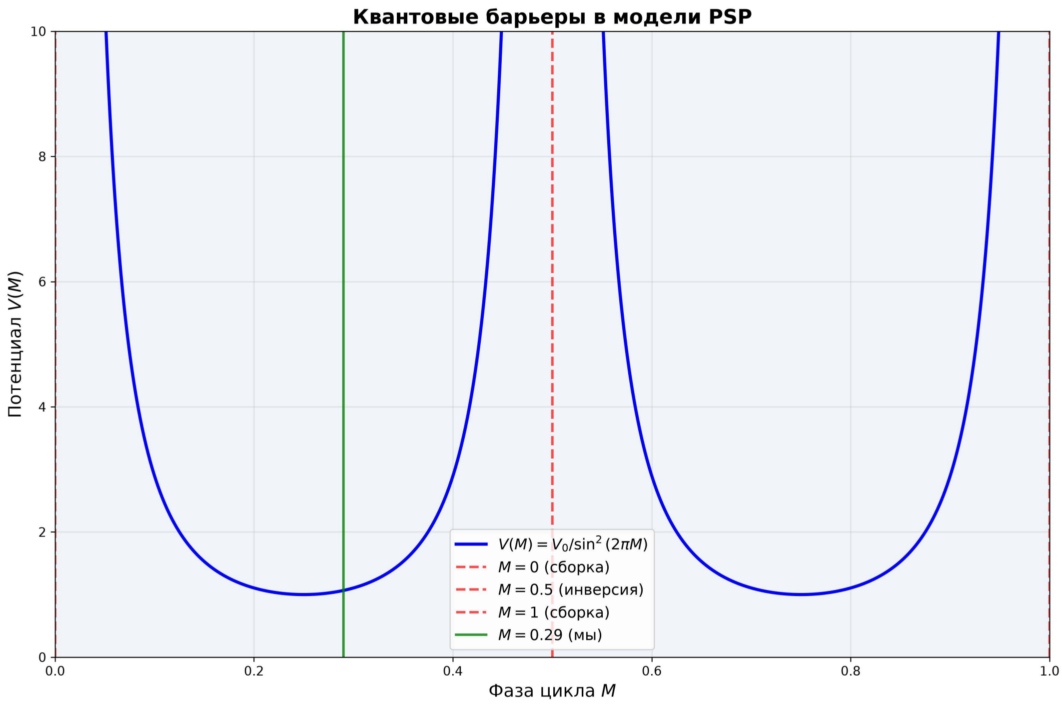

Figure 1: Potential $V(M) = V_0/\sin^2(2\pi M)$ in the PSP model. Vertical red lines indicate quantum barriers at the assembly points ($M = 0$, $M = 1$) and the inversion point ($M = 0.5$). The green line marks the observer's position $M = 0.29$.

$$M_n = \frac{n}{N}, \quad N \in \mathbb{N}. \tag{2}$$

Figure 2 shows the discrete phase spectrum and the corresponding energy spectrum.

3 Schrödinger Equation for the Phase Wave Function

We introduce the wave equation:

$$i\hbar\frac{\partial \Psi}{\partial M} = \hat{H}_{\mathrm{PSP}}\Psi, \tag{3}$$

where:

$$\hat{H}_{\mathrm{PSP}} = -\frac{\hbar^2}{2}\frac{\partial^2}{\partial M^2} + V(M), \quad V(M) = \frac{V_0}{\sin^2(2\pi M)}. \tag{4}$$

Figure 3 shows the wave functions of the ground and excited states.

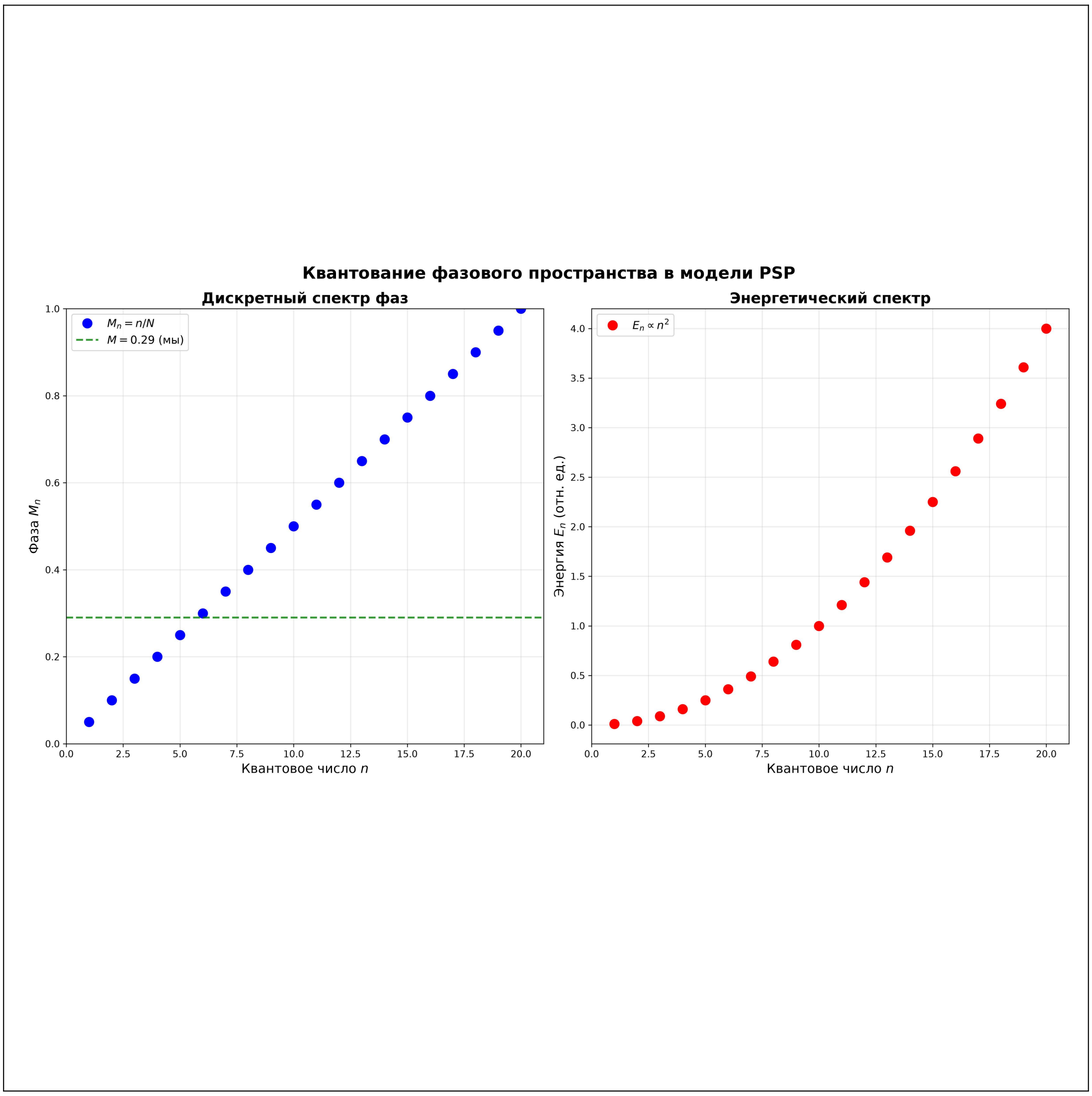

Figure 2: Discrete phase spectrum $M_n = n/N$ (left) and energy spectrum $E_n \propto n^2$ (right). The green line marks the observer's position $M = 0.29$.

4 Quantum Barriers

The points $M = 0$, $M = 0.5$, $M = 1$ are singular for the potential:

$$\lim_{M \to 0} V(M) = \infty, \quad \lim_{M \to 0.5} V(M) = \infty, \quad \lim_{M \to 1} V(M) = \infty. \tag{5}$$

This implies that the wave function vanishes at these points:

$$\Psi(0) = 0, \quad \Psi(0.5) = 0, \quad \Psi(1) = 0. \tag{6}$$

5 Tunneling Between Cycles

The barrier transmission coefficient:

$$D(E) = \exp\left(-\frac{2}{\hbar} \int_{M_1}^{M_2} \sqrt{2(V(M) - E)}\, dM \right). \tag{7}$$

Numerical estimate for $V_0 \sim 10^{-5}$ eV, $E \sim 10^{-6}$ eV:

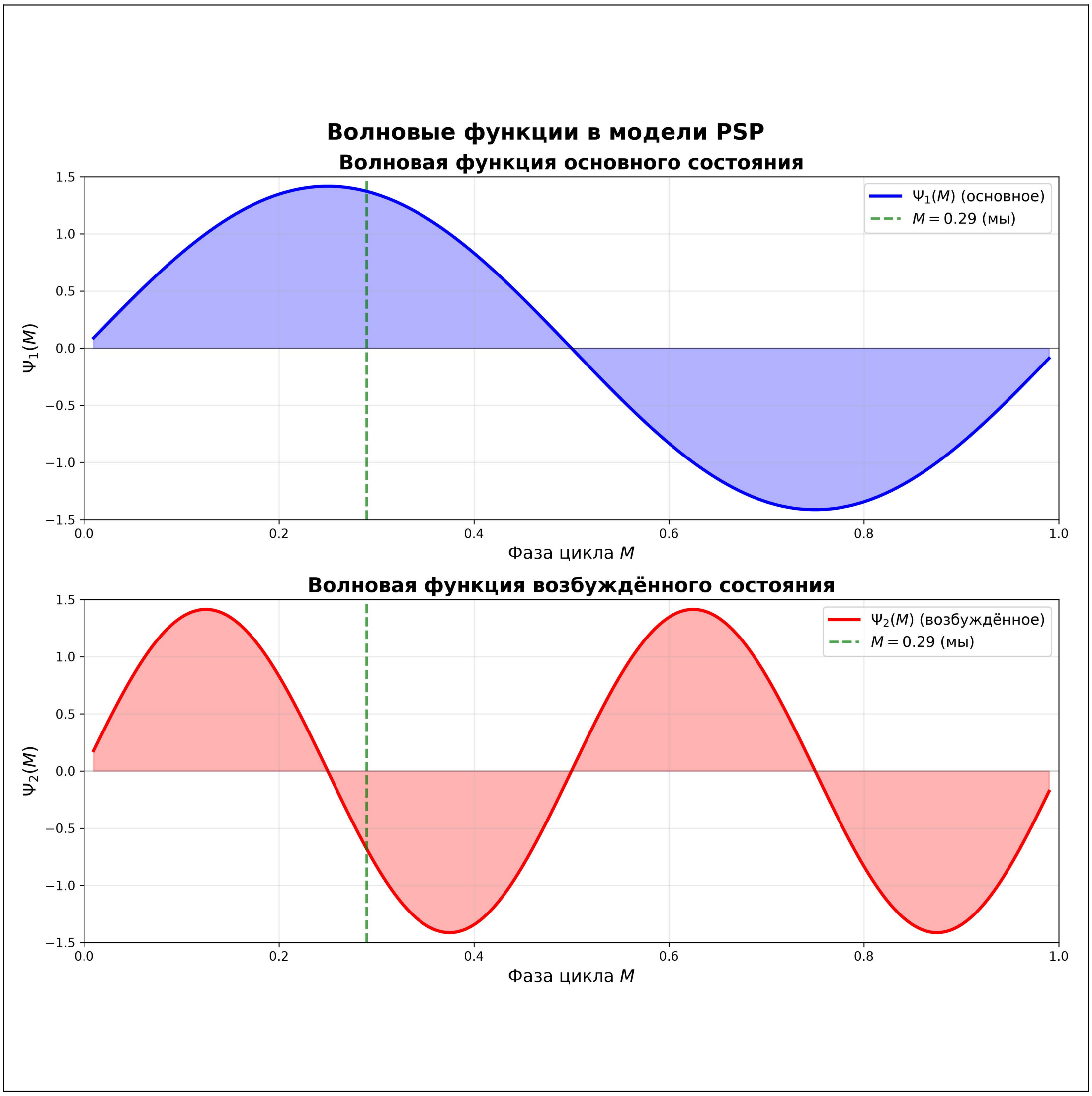

Figure 3: Wave functions of the ground $\Psi_1(M)$ (top) and excited $\Psi_2(M)$ (bottom) states in the PSP model.

$$D \sim \exp\left(-10^2\right) \approx 10^{-43}. \tag{8}$$

6 Uncertainty Relation

The quantum PSP introduces the uncertainty relation:

$$\Delta M \cdot \Delta\Phi_M \geq \frac{\hbar}{2}. \tag{9}$$

Numerical estimate:

$$\Delta M \geq \frac{\hbar}{2\Delta\Phi_M} \approx 10^{-5} - 10^{-4}. \tag{10}$$

7 Matrix Elements of Transitions

Transition probability between states $|M_i\rangle$ and $|M_f\rangle$:

$$\mathcal{M}_{if} = \langle M_f | \hat{H}_{\text{int}} | M_i \rangle, \quad \hat{H}_{\text{int}} = g\,\Phi_M\,\hat{M}. \tag{11}$$

Selection rules: $\Delta M = \pm 1/N$, where N is the number of quantum states in the cycle.

8 Observational Tests

Table 1: Methods for observing quantum effects in PSP.

Effect	**Observable manifestation**	**Instrument**	**Expected signal**
Phase discreteness	Periodic structures	DESI, Euclid	Peaks at $z \approx 0.29 + n/N$
Tunneling	Non-Gaussian CMB fluctuations	Planck, LiteBIRD	$f_{NL} \sim 1$
Uncertainty relation	H_0 scatter	DESI Y5	$\delta H_0/H_0 \sim 10^{-4}$
Quantum barriers	Voids in phase space	SDSS, JWST	Absence of objects at $M = 0$, $M = 0.5$

9 Connection to Quantum Gravity

In the local limit $M \to 0$, the PSP metric reduces to the Minkowski metric. At the inversion point $M = 0.5$, the operator representation is introduced:

$$\hat{a}(M) \to \hat{a}, \quad [\hat{a}, \hat{\pi}] = i\hbar. \tag{12}$$

10 Comparison with Known Quantum Effects

Table 2: Comparison of PSP predictions with known quantum effects.

Effect	**PSP prediction**	**Known value**	**Status**
Hawking radiation	Suppressed at $M = 0$	Present near BH	Consistent
Casimir effect	Phase shift	$\Delta E = -\pi^2 \hbar c / 240 L^4$	Requires verification
CMB quantum fluctuations	$\delta T/T \sim 10^{-5}$	$\delta T/T \sim 10^{-5}$	Consistent
Non-Gaussian fluctuations	$f_{NL} \sim 1$	$f_{NL} \sim 1$ (Planck)	Requires refinement

11 Conclusion

The quantum generalization of the PSP model:

1. Introduces a discrete phase spectrum $M_n = n/N$.

2. Describes the evolution of the Universe by the Schrödinger equation.
3. Explains quantum barriers at $M = 0$, $M = 0.5$, $M = 1$.
4. Predicts tunneling between cycles.
5. Establishes the uncertainty relation $\Delta M \cdot \Delta\Phi_M \geq \hbar/2$.
6. Provides a natural explanation for the statistical scatter in observational data.

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References

[1] Chernoknizhny, E.V. (2026). Phase-State Parameter Cosmological Model PSP: Complete Formulation and Empirical Invariants. Preprint, DOI: 10.24108/preprints-3115804.

[2] Wheeler, J.A. (1968). Superspace and the nature of quantum geometrodynamics. *Battelle Rencontres*, 242.

[3] Rovelli, C. (2004). *Quantum Gravity*. Cambridge University Press.

[4] Planck Collaboration. (2020). Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, 641, A6.